

PROFESSIONAL SUMMARY & TECHNICAL PROFICIENCIES

- Biochemical Engineer/Scientist with expertise in both upstream and downstream bioprocessing, specializing in **human and animal cell culture, viral vector production and purification, and downstream purification techniques**
- **Completed and led multiple industry-academia collaboration project** with strong cross-functional work and communication skills with an ability to manage dedicated projects and coordinate design and practical group
- Strong expertise of **downstream purification processes**: virus inactivation, Protein A chromatography, virus filtration, TFF, UF/DF, AEX, affinity chromatography, and process intensification strategies
- Experienced with **protein and material analysis & characterization**: DLS, FTIR, AFM, HPLC, ELISA, PCR, FPLC, LC-MS, NMR, HIC, and virus infectivity assays
- Skilled in **advanced instrumentation** including AKTA Avant/Pure, SCIEX PA 800 Plus, Pendotech NFF/TFF systems, confocal microscopy, TEM, DLS (DelsaNano), Optima XPN-100 ultracentrifuge, and SEC
- Document, analyze, and interpret results in **ELN and LIMS systems**.
- **Bilingual**: Fluent in Chinese (Native) and Fluent in English (Proficient in Business)

EDUCATION

University of Arkansas, Fayetteville, AR, United State Sep 2019- May 2025
Doctor of Philosophy in Biomedical Engineering, Research focuses on biochemical and biotechnology

University of Arkansas, Fayetteville, AR, United State Sep 2015 - May 2019
Bachelor of Science in Biomedical Engineering, Honors research project on downstream virus clearance

PROFESSIONAL EXPERIENCE

Plumino Precision Fermentation USA, Inc., Cape Girardeau, MO Nov 2025 - present
Process Engineer

- **Optimized industrial fermentation and downstream purification processes** for amino acid manufacturing, applying biochemical engineering principles to improve process efficiency, product quality, and manufacturing reliability.
- **Analyzed production performance** using mass and energy balances, transport phenomena, separation science, and process data to identify opportunities for raw material reduction, utility optimization, wastewater reduction, and cost savings.
- **Led engineering-based troubleshooting and root cause analysis** for process deviations, equipment performance issues, and wastewater-related challenges, developing corrective and preventive actions grounded in technical data.
- **Prepared technical reports, feasibility analyses**, and process improvement recommendations for senior management to support plant optimization, process robustness, and long-term manufacturing efficiency.
- **Supported SAP implementation and manufacturing digitalization initiatives**, including blueprint design, production management process mapping, production workflow standardization, and system requirement definition.
- **Designed SAP-aligned production management processes** covering material flow, batch production logic, production orders, inventory movements, process traceability, and finished product management.

Membrane Research Laboratory, University of Arkansas Jan 2020 - May 2025
Graduate Research Assistant

- **Advanced process innovation and intensification studies**, optimizing filtration and chromatography operations for improved efficiency, robustness, and scalability; evaluated critical parameters (flux, throughput, fouling behavior, viral clearance, protein recovery) to advance next-generation downstream processes.
- **Extensive mammalian cell culture experience** with suspension/adherent lines (CHO, HEK, NB324K, PG-4, A9) and virus infectivity assays (TCID₅₀, fluorescence-based, flow cytometry, TEM, PCR).

- **Fabricated electrospun membranes** with UV-initiated GMA polymerization; **characterized membranes** (zeta potential, static/dynamic contact angle, porosity, flux, SEM).
- **Experienced in E. coli cultivation and plasmid extraction** (alkaline lysis/kit-based methods) for vector amplification and molecular cloning, with proficiency in aseptic techniques, bacterial culture, and DNA handling, and a strong understanding of plasmid-based gene expression in metabolic engineering.
- **Investigated downstream operations**, including clarification, depth/microfiltration, ion-exchange chromatography, virus filtration, UF/DF for process innovation, parameter optimization, and scalability testing.
- **Performed protein characterization** using analytical methods (HPLC, SDS-PAGE, ELISA, UV absorbance), assessing purity, yield, structural integrity, and recovery to support process optimization and product quality evaluation.
- **Applied analytical modeling** of flux decline, fouling kinetics, and virus retention to interpret data and guide process optimization.
- **Ensured viral safety and regulatory alignment** by achieving log reduction values (LRV) consistent with industry and GMP expectations.
- **Collaborated with industry supervisors** through presentations and technical reports; developed SOPs to support knowledge transfer and project alignment.
- **Presented research at international conferences** and contributed to peer-reviewed publications on virus filtration and downstream process development.
- **Provide virus infectivity assay support** for virus clearance studies run for all research projects in the team and training junior graduate students in laboratory techniques.

AbbVie Inc. North Chicago, United States
Process Engineering Intern

Sep 2024 - Jan 2025

- **Sustainable Process Innovation:** Investigate the application of pervaporation to replace conventional distillations to reduce the solvent consumption and subsequent waste generation significantly
- **Parameter Optimization:** Investigated the impact of inert gas sweep, feed pressurization, temperature variation, and co-solvents on water activity and overall water flux in pervaporation
- **Scale-up Analysis:** analyzed experimental results to evaluate scale up potential and assess the practical implementation of pervaporation processes in industrial settings

PROJECT & RESEARCH

Optimization and Characterization of Lentivirus Production	Feb 2025 - May 2025
• Conducted optimization of high-yield lentiviral vector production, including plasmid extraction, ultracentrifugation, and affinity chromatography-based purification • Characterized and quantified lentivirus using SDS-PAGE, PicoGreen assay, TEM, qPCR, and infectivity assays to assess vector quality and functional titer	
Evaluation of xMuLV Clearance in AAV Purification Process	April 2024 - August 2024
• Developed SOPs for in-house production and purification of high-titer xenotropic murine leukemia virus (xMuLV) and adeno-associated virus (AAV), supporting virus clearance validation in downstream purification • Evaluated xMuLV clearance using Planova 35N virus filters under defined AAV feed conditions at constant pressure; virus removal quantified by infectivity assay and qPCR, and AAV integrity assessed via PicoGreen assay, ELISA, and TEM	
Advanced Strategies for Investigating Underlying Virus Filtration Mechanisms	Fall 2022 - April 2024
• Developed standardized procedures for in-house production and purification of high-titer minute virus of mice (MVM); evaluated virus retention performance of three commercial virus filters under varying flow rates, pressures, and feed compositions in both constant pressure and constant flux modes • Applied confocal microscopy to visualize MVM retention within membrane structures; quantified virus clearance and protein recovery to assess filtration efficiency and mechanistic differences across filters	
Virus Clearance (MVM) Assessment and Comparison using Different AEX Chromatography	Fall 2019-Fall 2020
• Evaluated MVM clearance using BSA and different types of mAbs under defined feed conditions across various commercial AEX membranes; compared virus removal performance of membrane-based and traditional resin-based AEX systems at different mAb concentrations and flow rates	

- Assessed and contrasted Q membrane performance with a multimodal membrane for virus clearance efficiency, focusing on the influence of mAb concentration and binding modality

INVOLVEMENT & LEADERSHIP

-
- | | |
|--|-------------------------|
| • NSF Non-Academic Research Internships Program | Fall 2024 – Spring 2025 |
| • BMEG Community & Student Engagement Fellow | Fall 2023 – Fall 2024 |
| • Member of American Institute of Chemical Engineers (AIChE) | Fall 2022 - Spring 2025 |
| • Vice President, BMEG Graduate Student Association University of Arkansas | Fall 2022 – Fall 2023 |
| • College of Engineering at-Large Representative/DEI Committee Co-Chair (GPSC) | Fall 2022 – Spring 2024 |
| • Member of American Association of Pharmaceuticals Scientists (AAPS) | Spring 2023 – Fall 2025 |
| • Treasurer, BMEG Graduate Student Association University of Arkansas | Fall 2020 - Fall 2022 |

PUBLICATION and CONFERENCE PRESENTATION

-
- Xu, W.,** Qian, X., Shirataki, H., & Wickramasinghe, S. R. (2026). Modeling the effect of monoclonal antibody on the performance of small virus filters using an advanced multilayer membrane model. *Journal of Membrane Science*, 125475.
 - Xu, W.,** Qian, X., Shirataki, H., Strauss, D., & Wickramasinghe, R. (2025). The effects of membrane structure and filtration conditions on virus filtration. *Journal of Membrane Science*, 124781.
 - Xu, W.,** Qian, X., Shirataki, H., Straus, D., & Wickramasinghe, S. R. (2025). Visualizing the Effect of Process Pause on Virus Entrapment During Constant Flux Virus Filtration. *Membranes*, 16(1), 6.
 - Xu, W.,** Qian, X., Shirataki, H., Strauss, D., & Wickramasinghe, R. (2025). Optimization and Scale-Up of Pervaporation for Dehydration of 2-MeTHF: An enabling technology for pharmaceutical processing. *Separation and purification technology*. (submitted; under review)
 - Chen, S. T., **Xu, W.**, Cai, K., Ferreira, G., Wickramasinghe, S. R., & Qian, X. (2022). Factors affecting robustness of anion exchange chromatography: Selective retention of minute virus of mice using membrane media. *Journal of Chromatography B*, 1210, 123449.
 - Xiao, Y., Tian, Y., **Xu, W.**, & Zhu, J. (2024). Photodegradation of microplastics through nanomaterials: Insights into photocatalysts modification and detailed mechanisms. *Materials*, 17(11), 2755.